



Classification of GAIT data using machine learning technique to categorize human locomotion

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Outline

Problem Statement/Proposed Solution

Aim

Motivation

Model Description

Proposed work and Methodology

Data Collection

Classification of gait data using KNN, K-Mean, Back Propagation

Result analysis, Conclusion, Future Scope and Limitation

My Publications (4 conferences+2 Journals+ 5 e-papers) & References

What is GAIT?

Gait

- **Gait refers to the manner in which a person walks, and is one of the few biometric traits that can be used to recognize people at a distance.**
- **Therefore, this trait is very appropriate in surveillance scenarios.**
- **Hence, the selection of a good model to represent the human body is pivotal to the efficient functioning of a gait recognition system.**
- **However, the gait of an individual is affected by footwear, clothing, affliction of the legs, or walking surface.**

Human gait

- Human gait analysis refers to the study of human locomotion through the evaluation of kinematic, kinetic and spatial/temporal parameters.
- This field of study finds application in a large variety of domains, including entertainment, healthcare and pedestrian navigation.
- The act of walking involves the coordination of different human body parts, such as the skeleton, the muscle and the neural systems.
- Various factors can affect the complex interaction between the body parts, for example any pathological nature will require a distinction between — **normal gait** and **pathological gait**.
- Normal gait refers to the naturalistic and general human walking parameters without differentiating the age, sex or individual physical parameters.
- Conversely, —pathological gaitll refers to an abnormal gait, for example affected by pathologies such as muscle weakness or skeleton deformities.

Problem Statement

- Identifying normal or pathological human gait poses many challenges. Therefore a robust method for detecting and capturing the body, face and/or gait of each individual person has to be found.
- Traits or features have to be extracted from each detection. How many and which features are needed for classification? Which classification method gives the most accurate results? How high will the classification rate be?

Proposed Solution

We have developed the computational model for prediction, formal verification and analyses of bipedal locomotion using machine learning technique for modelling .

AIM



- The aim of this masters thesis is to evaluate and implement a sufficiently robust and accurate method for human to identify whether it's a **normal gait** or **pathological gait** and to improve the speed and accuracy of classification in the presence of category hierarchies.

Motivation

Motivation

*compatible with human
environment*

understand what causes
humanoids to *fall*

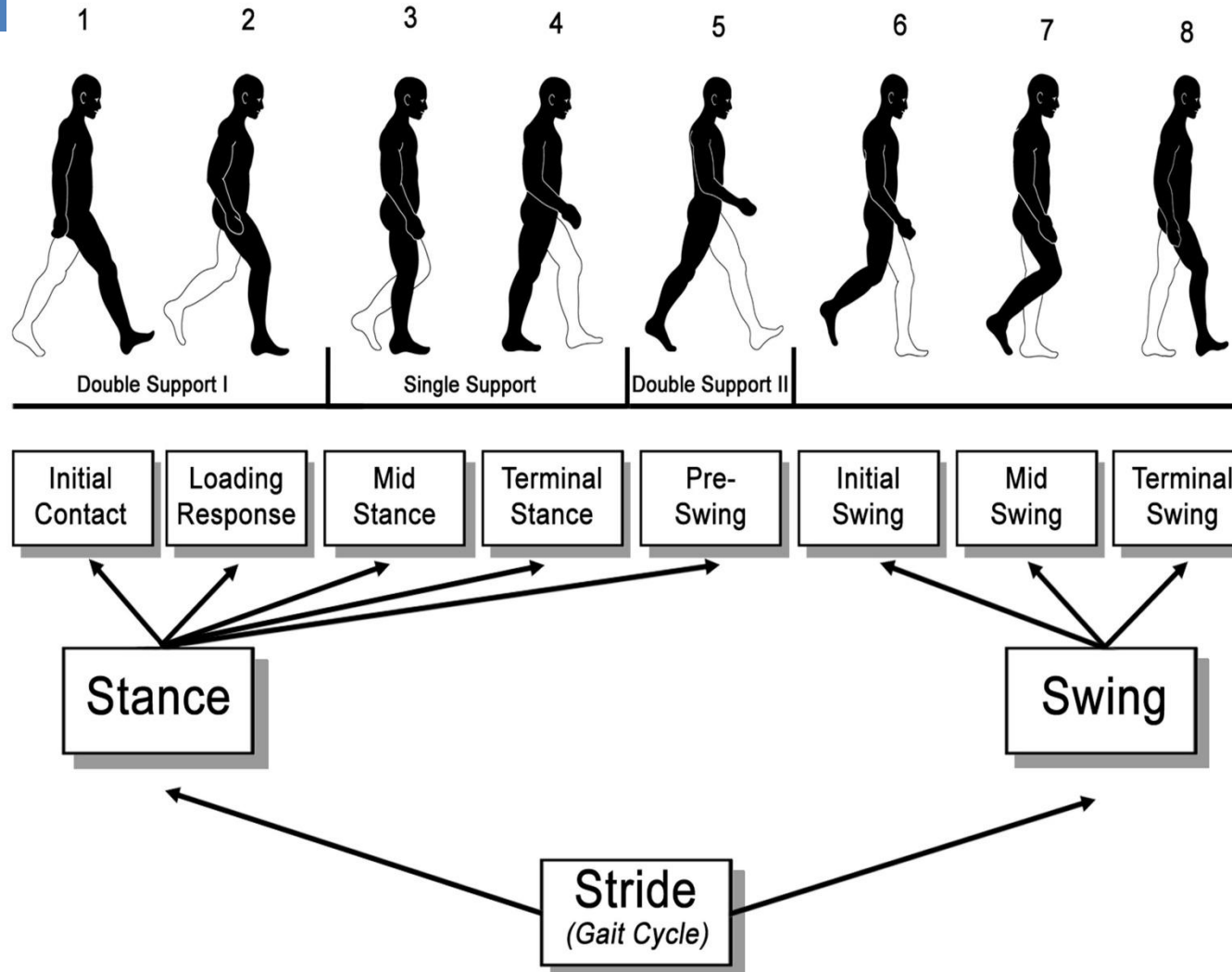
help *elderly or persons
with disabilities*

systematic investigation of *complex biped locomotion
dynamics*

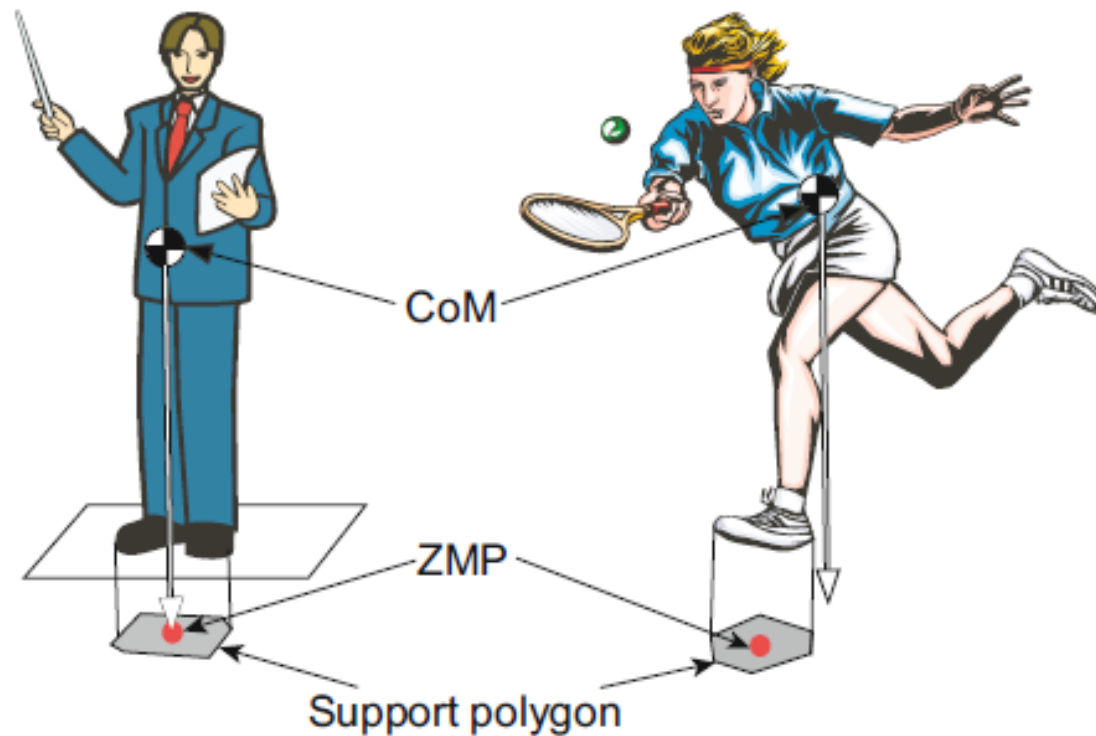
focused only on the
*locomotion on flat
surfaces.*

Human Gait Biomechanics

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1. *Static walk*
2. *Dynamic walk*



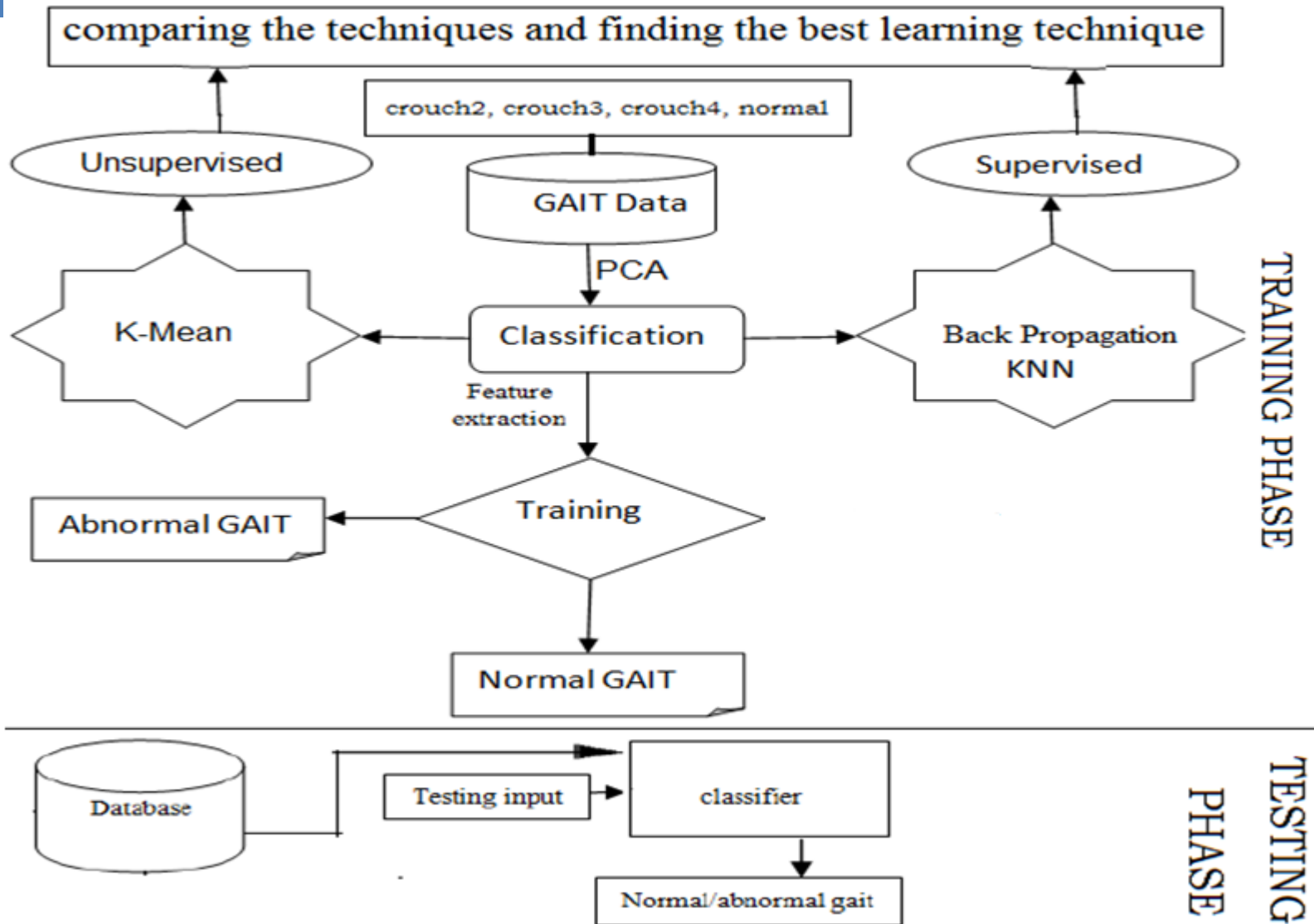
(a) A standing human

(b) A human in action

CoG, ZMP, and Support Polygon

Proposed Work

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METHODOLOGY

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METHODOLOGY

- Firstly we collect the GAIT data then features are extracted and we classify the GAIT dataset.
- Training and testing of system is done.
- After the classification is done result is shown as normal or pathological GAIT.
- Then 10 fold cross-validation test also conducted to validate the statistical significance of the results.

Phase Wise Development

Study and Data Collection of bipedal walk and features are extracted

Training and testing of system is done

Human Push Recovery data Classification using Neural network, machine learning

Then 10 fold cross-validation test also conducted to validate the statistical significance of the results

Data Collection of bipedal walk

Step 1

- Acquiring human gait data
- (HMCD/OpenSim)

Step 2

- Analyzing walking behavior and constraints
- General form of data representation

Step 3

- Domain Breakdown of Gait Cycle
- Formal framework to model

Step 4

- Classification
 - KNN, Neural network (back propagation)
 - K-means

Step 5

- Verification by Simulation in OpenSim/Matlab
 - Lower Extremity/Biped
 - Generic Model

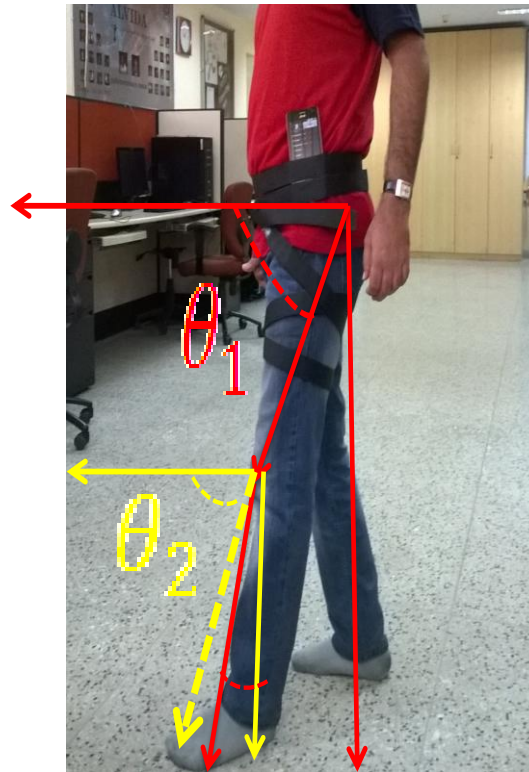
Data Collection

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□ HMDC

- To capture the human data we use HMCD. From this device different joint angles (knee, hip, and ankle) are captured to get the human GAIT data.

Empirical Study of Gait by HMCD



Methodology -Preprocessing - Real data collection and Zero correction

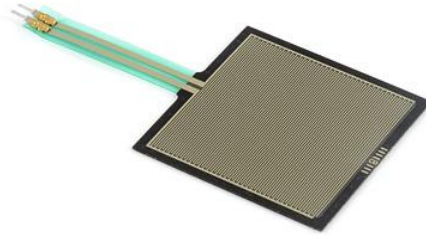
1. Real data acquisition by HMCD suit. HMCD consist of:-
 - o Potentiometer
 - o Force Sensor (FSR)
 - o Accelerometer
2. Study the behavior of data and also Processing the data.
 - o Zero correction.
 - o Applying respective formula.
3. Plotting graph all the six joint and COM with respect to force.
 - o Curve fitting using interpolation.
 - o Analysis of nature and behavior of graphs.

$$\text{angle(in degree)} = [(\theta - \theta_0) * \frac{300}{1000}]$$

$$\text{Force (Newton)} = f * (9.8 / 1000)$$

Real data collection-Hardware used-

HMCD (Human Motion Capture Device) Suit- Consist of following

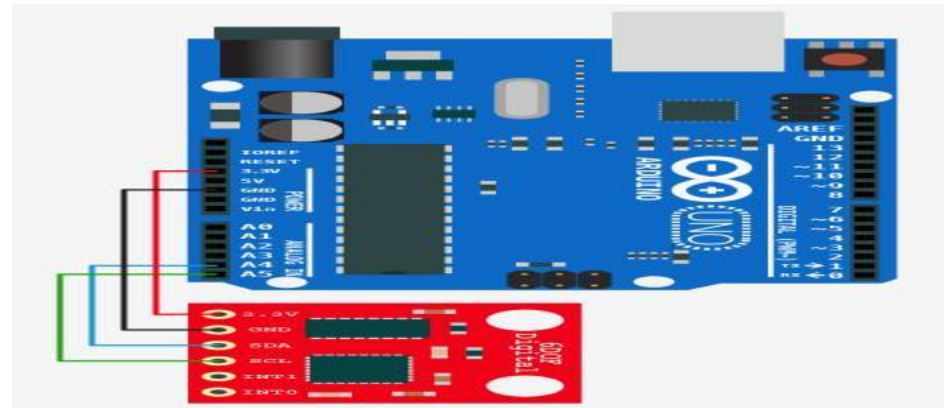


☐ Rotation sensors 1109

☐ The rotation sensor can rotate only 0 to 300 degree.

► Force sensor:

- ☐ Force sensor that is used is known as FSR 3105 (Force sensing resistor).
- ☐ This force sensor gives values in digital count and the range of digital count is 1N to 100N and the sensing area of FSR 3105 is 14.5 cm².



Phase by Phase Study & Development- Data Acquisition By HMCD

Subject in indigenous developed HMCD wearable suit



HMCD suit with Accelerometer HMCD suit with force sensor

Closed eye without lunging video

Closed eye with lunging video

Experimental Setup for Humanoid Push recovery – Frontal /Backward Plane

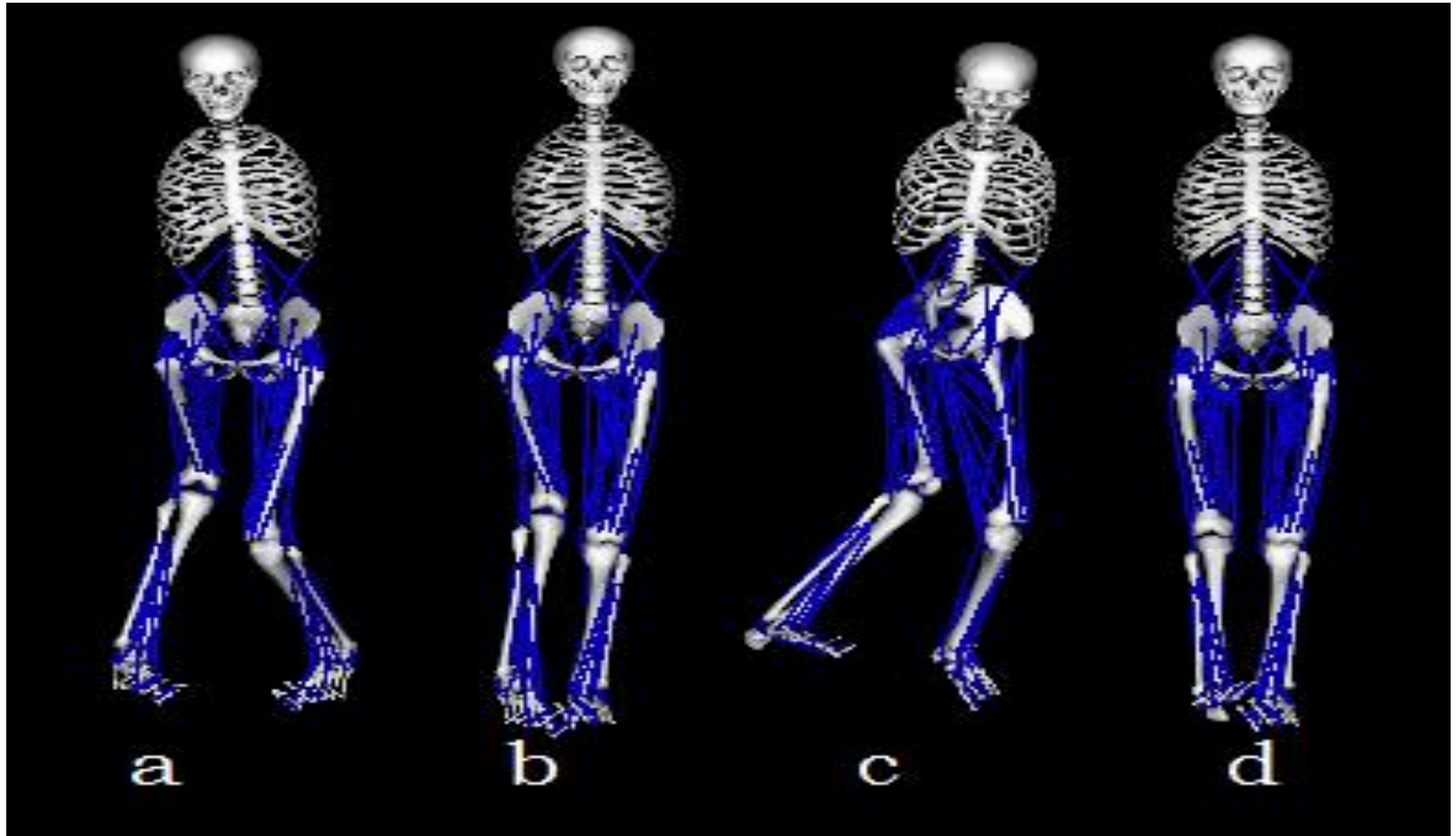
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Assumptions

- ❑ The motion of arms is neglected
- ❑ Lesser degrees of freedom are assumed:.
- ❑ Floor is assumed to be flat.
- ❑ Rugged terrain is not considered. Friction is also neglected.

Simulation of crouch data in openSim

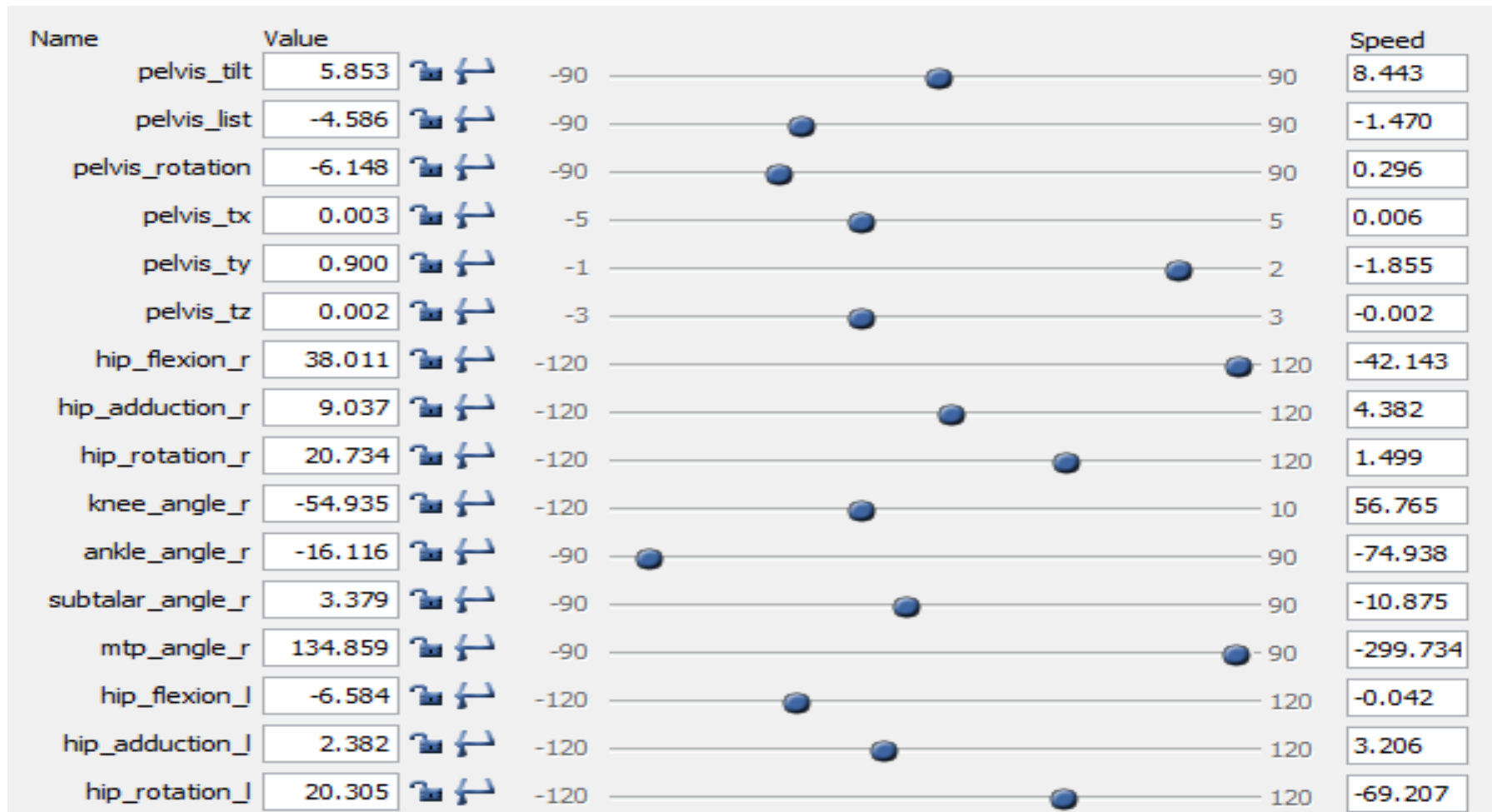
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Values of crouch2 dataset

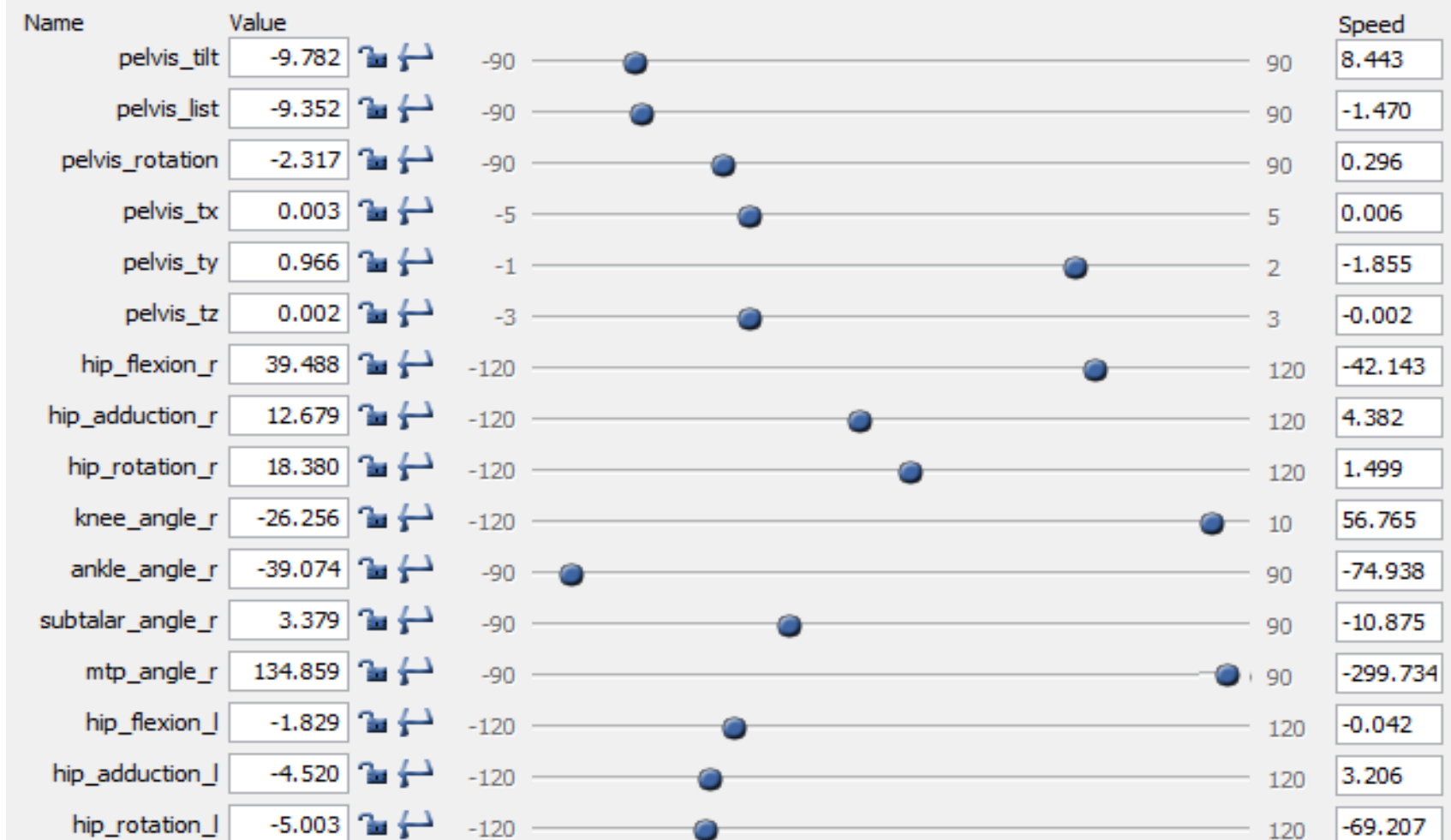
22



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Values of crouch3 dataset

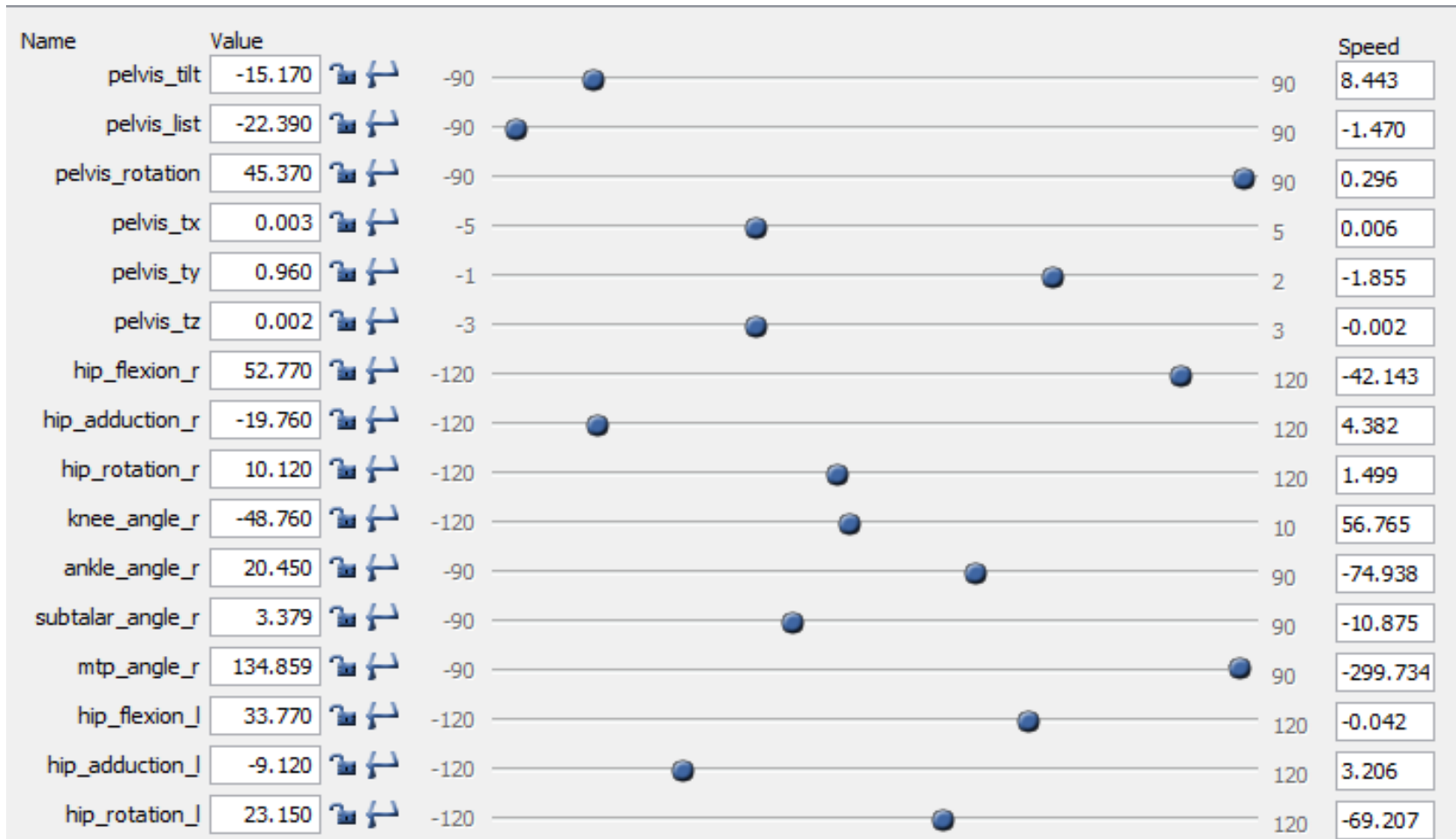
23



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Values of crouch4 dataset

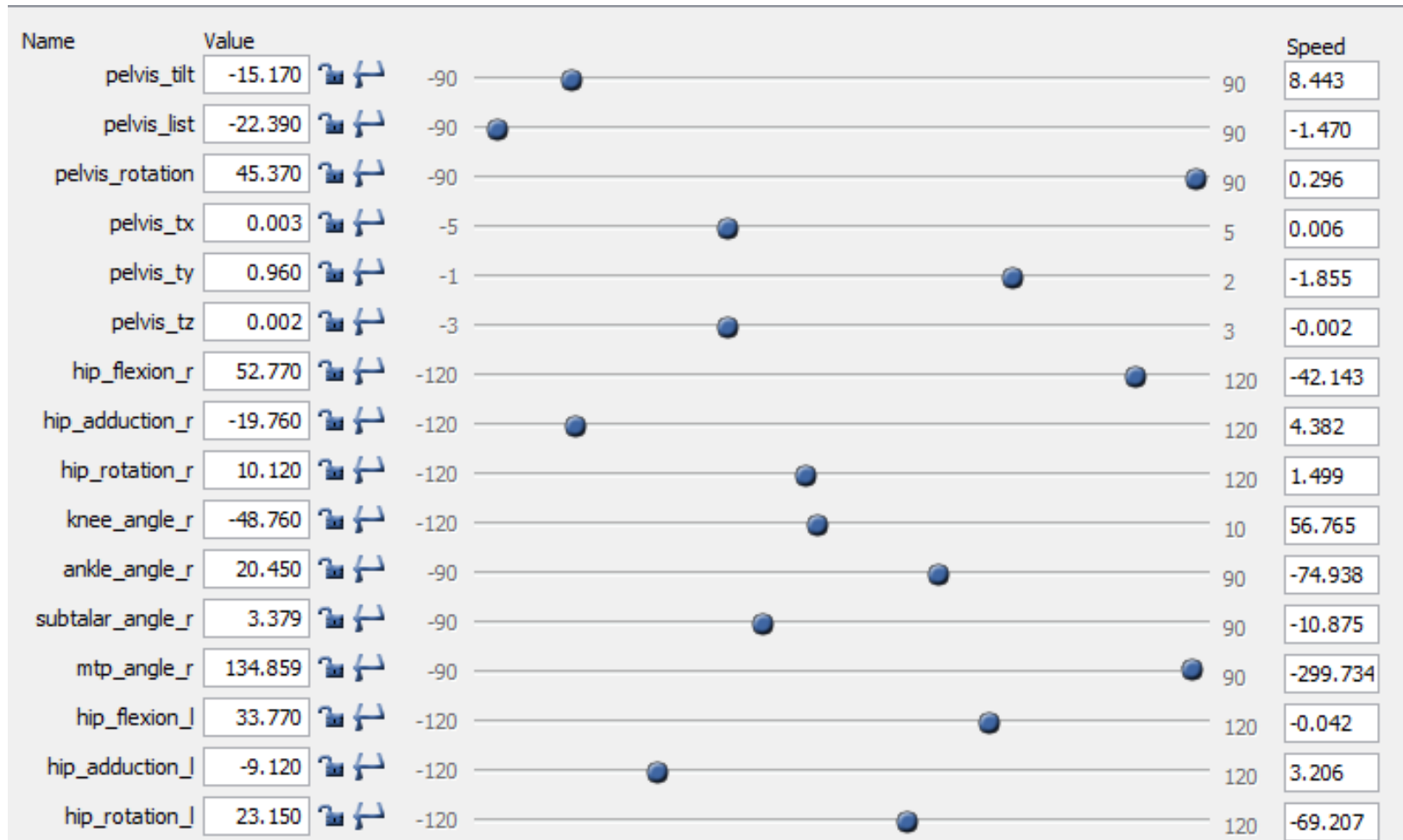
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Values of normal dataset

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Analysis

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When the is simulated, then the coordinates of the human GAIT data set is captured and shown in figure.4, 5, 6, 7 for crouch2, crouch3, crouch4, normal datasets respectively.

Plot of crouch2, crouch3, crouch4, and normal datasets respectively.

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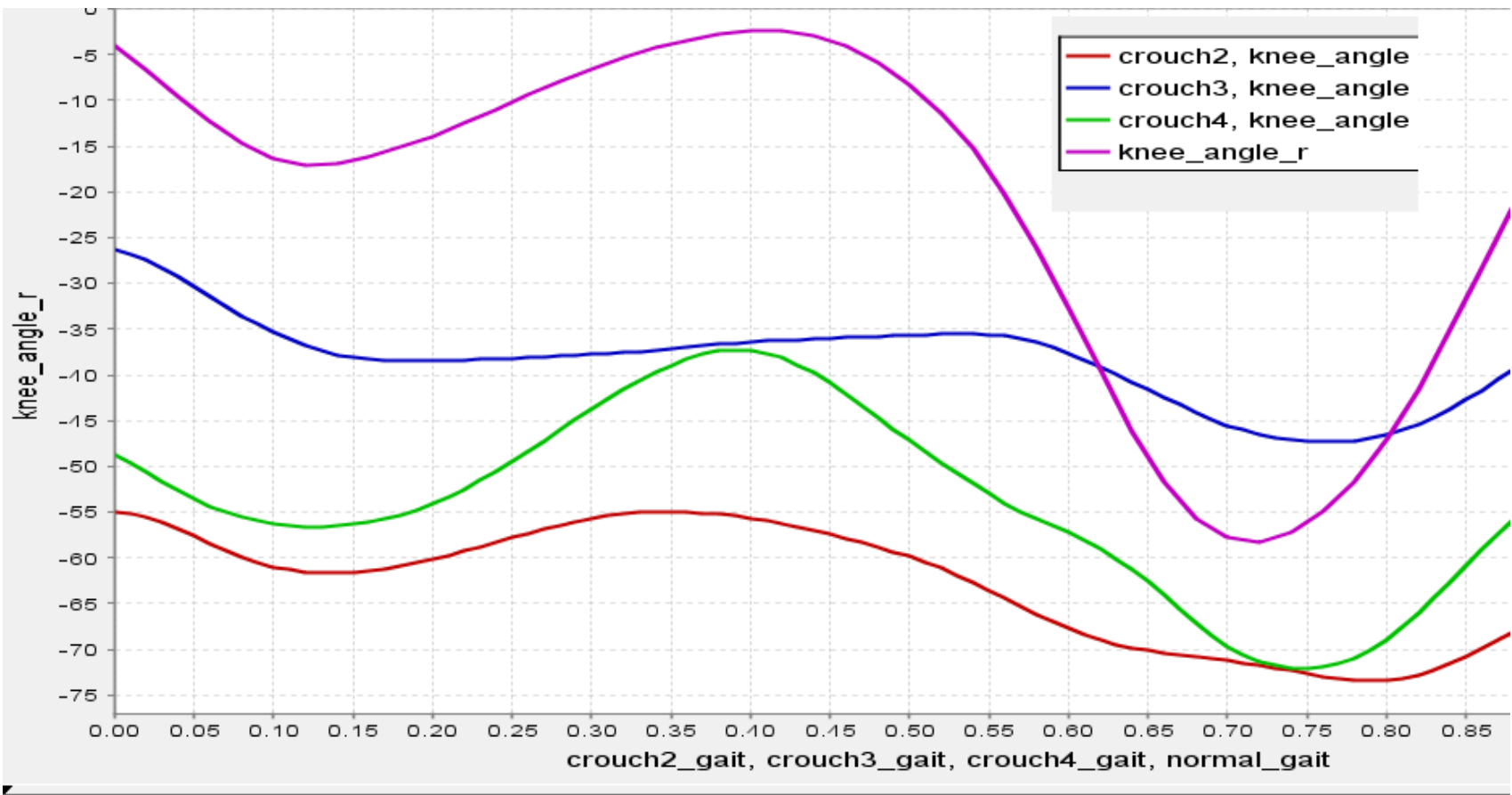


Figure shows plot of knee angle curve of data set crouch2 (red), crouch3 (blue), and crouch4 (green), normal (purple) respectively.

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Feature Selection

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Feature Category	Feature Name
F1	Pelvis_tilt
F2	Pelvis_list
F3	Pelvis_rotation
F4	Pelvis_tx
F5	Pelvis_ty
F6	Pelvis_tz
F7	hip_flexion_r
F8	hip_adduction_r
F9	hip_rotation_r
F10	knee_angle_r
F11	ankle_angle_r
F12	ankle_angle_l
F13	knee_angle_l
F14	hip_flexion_l
F15	hip_adduction_l
F16	hip_rotation_l

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CLASSIFICATION

KNN, BP, K-Mean

Classification

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Classification is a collection of different methods for categorizing data. The problem of classification can be illustrated in a scatter plot as seen in figure 9. To correctly classify the data one wishes to find features that separate each group of data into clusters. Before classification it is common practice to normalize the data. Doing so will make the distances in different directions to carry the same weight, i.e. assuming that all features have the same importance.

Classification

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- Classification of GAIT data is done through two following techniques:
- KNN
- BP
- K-Mean

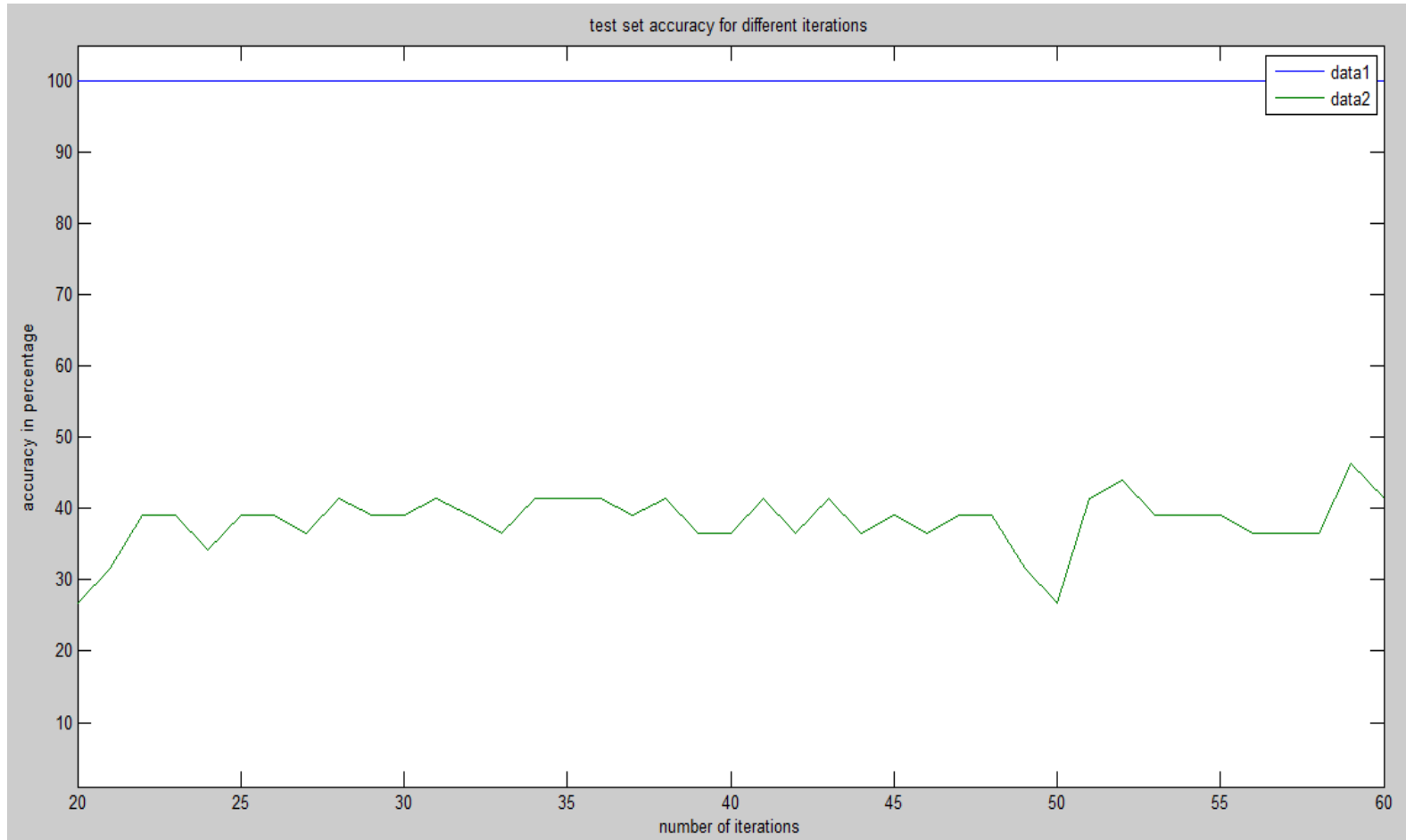
Neural Network (Back Propagation)

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- Back propagation, an abbreviation for "backward propagation of errors", is a common method of training artificial neural networks used in conjunction with an optimization method such as gradient descent. The method calculates the gradient of a loss function with respect to all the weights in the network. The gradient is fed to the optimization method which in turn uses it to update the weights, in an attempt to minimize the loss function.

Result

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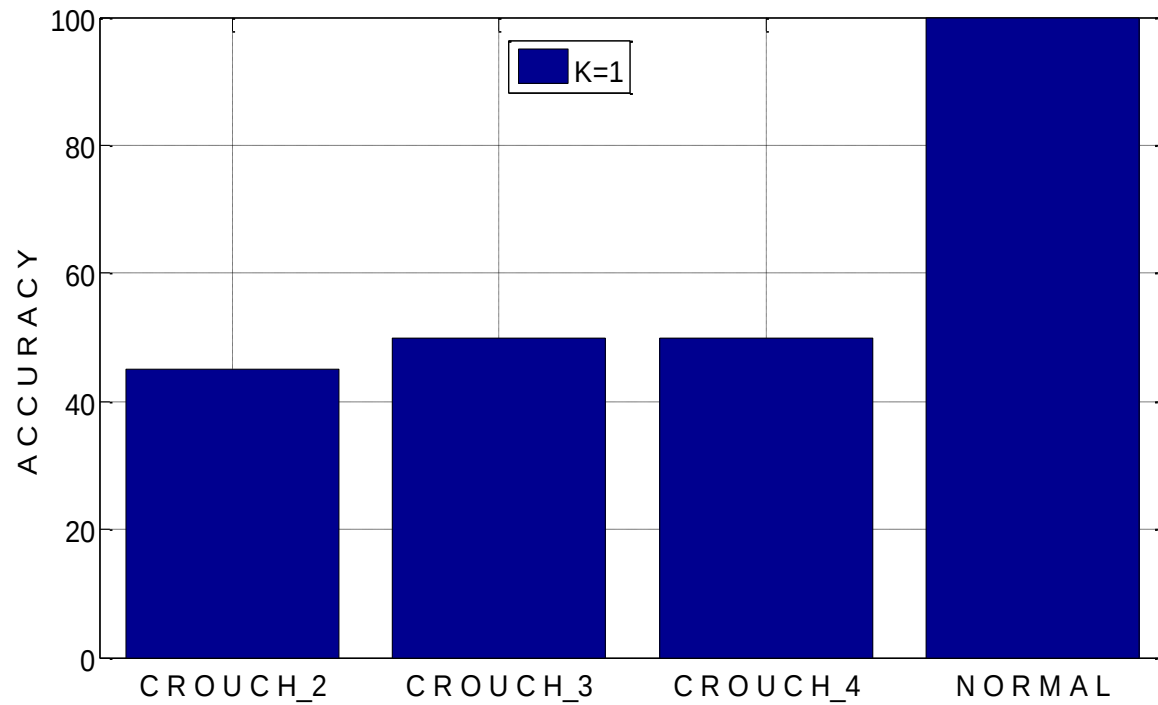
KNN

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- K-Nearest Neighbours algorithm (or k-NN for short) is a non-parametric method used for classification. In this, the input consists of the k closest training examples in the feature space. The output depends on whether k-NN is used for classification:
- In k-NN classification, the output is a class membership. An object is classified by a majority vote of its neighbours, with the object being assigned to the class most common among its k nearest neighbours (k is a positive integer, typically small). If $k = 1$, then the object is simply assigned to the class of that single nearest neighbour.

KNN Result

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K-Mean

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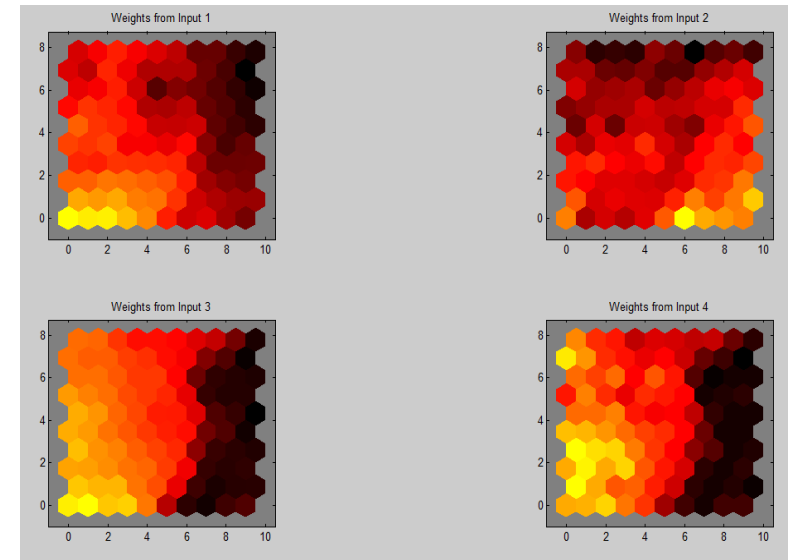
- ❑ The K-Means algorithm is a method to cluster objects based on their attributes into k partitions.
- ❑ It assumes that the k clusters exhibit Gaussian distributions.
- ❑ It assumes that the object attributes form a vector space.
- ❑ The objective it tries to achieve is to minimize total intra-cluster variance.

K-Mean result

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Validation

Correlation coefficient	0.9955
Mean absolute error	8.2394
Root mean squared error	117.3938
Relative absolute error	0.741 %
Root relative squared error	9.4381 %
Total Number of Instances	203



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Overall result

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	ACCURACY COMPARISON FOR BACK PROPAGATION, K-MEANS AND KNN		
	<i>Back Propagation</i>	<i>KNN</i>	<i>K-Mean</i>
Accuraccy (Overall)	69.825%	61.25%	50%

Accuracy	Category/Method	K-Means	NN(BP)	KNN(k=1,2,3,4,5)
Training Accuracy		100%	100%	60%
Testing Accuracy	Normal Gait	54.68	56	100,100,100,100,100
Testing Accuracy	Crouch 2 Gait	51.43	43	50,50,50
Testing Accuracy	Crouch 3 Gait	33.19	43	50,50,50
Testing Accuracy	Crouch 4 Gait	51.80	53	45,40

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Conclusions

- GAIT analysis depends on age, sex, weigh, height i.e. based on human anatomy.
- The pattern of human corresponding knee and ankle joint angles are inversely proportional to each other.
- Knee is the most active joint in GAIT analysis of human.
- We analyzed the gait pattern of human that left side joints are more active than right side joints of a right handed person while in case of left handed person, right side joints are more active.

Conclusion

- Thanks to this strategy we could achieve 100% training accuracy of classification and 43% testing accuracy. For 16 input features the overall accuracy we get from back propagation algorithm is 69.825%
- Therefore the proposed system can be used to identify the human gait and to know which locomotion provides normal GAIT and abnormal GAIT.

Future work

- The gender of a subject can also affect the age classification. The future scope is the proposed system can be used in surveillance systems or to detect any upcoming disease related to GAIT human locomotion; reorganization system and testing accuracy of the system can be improved.
- The model can be used for classification of different cases of gait pattern, like healthy persons gait walk, crippled walking gait pattern etc. It can be used further for push recovery analysis.

Limitations:

- Essentially, for a given number of training samples, there is a limit for the maximum number of features, or the so-called dimensionality of the feature vector.
- With the growth of dimensionality (i.e., volume of the space), the number of required training samples must grow exponentially, otherwise, the data in high-dimensional space will become sparse.
- This scarcity of data can lead to over fitting since there are not enough training samples.

My Publications (4 conferences+2 Journals+ 5 e-papers)

SCI Journals (5): Total Impact Factor- 8.46, Research Gate Score-17.0

Journals

- 1 Anubha Parashar, Somya Goyal, Apoorva Parashar "Push Recovery for Humanoid Robot in Dynamic Environment and Classifying the Data Using K-Mean." in *International Journal of Interactive Multimedia and Artificial Intelligence (IJIMAI2016)*.
- 2. VS Bhaskar, AK Singh, J Dhruw, A Parashar, M Sharma, "Business and Social Behaviour Intelligence Analysis Using PSO." in *International Journal of Interactive Multimedia and Artificial Intelligence, (IJIMAI) Vol.2 (6), Page No. [69-74](#)*.

International Conferences

- 1. Anubha Parashar, Somya Goyal, Apoorva Parashar, Bharat Sahjalan, "Push Recovery for Humanoid Robot in Dynamic Environment and Classifying the Data Using K-Mean.", [Second International Conference on Information and Communication Technology for Competitive Strategies](#) (ICTCS-2016)(Conference Proceedings by **ACM** – ICPS ISBN No 978-1-4503-3962-9).
- 2. Anubha Parashar, Apoorva Parashar, Somya Goyal, " Classifying Gait Data using different machine learning techniques and finding the optimum technique of classification.", [International Conference on ICT for Sustainable Development](#) (ICT4SD-2016)(Conference Proceedings by **Springer** - Lecture Notes in Networks and Systems (LNNS Series)).
- 3. Anubha Parashar, Deepak Goyal, "Reorganisation of GAIT data using unsupervised learning ", *International Conference on Smart Trends for Information Technology and Computer Communications (SMARTCOM 2016)*.

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4. Anubha Parashar, Deepak Goyal, "Identification of GAIT data using machine learning technique to categorise human locomotion", *IEEE Conference at IIT KGP*.

E-Prints

1. A Parashar, S Kumar, VS Bhaskar, R Chinia, "Noisy Distance Measurements Using 3-D Localization with Rb-Rf Methods", available from <http://arxiv.org/ftp/arxiv/papers/1407/1407.2125.pdf>

2. A Parashar, V Bijalwan, "A Text to Speech Conversion Engine", available from

https://www.researchgate.net/profile/Vishwanath_Bijalwan3/publication/274195825_A_Text_to_Speech_Conversion_Engine/links/5518e2e70cf2d70ee27b752d.pdf

3. M Rani, A Parashar, J Chaturvedi, A Malviya, "Search Space Engine Optimize Search Using FCC_STF Algorithm in Fuzzy Co-Clustering", available from <http://arxiv.org/ftp/arxiv/papers/1407/1407.6952.pdf>

4. J Chaturvedi, A Parashar, AA Manjrekar, VS Bhaskar, "Social and Business Intelligence Analysis Using PSO", available from <http://arxiv.org/ftp/arxiv/papers/1407/1407.6090.pdf>

5. K Vaish, SM Rajesh, K Pasupatheeswaran, A Parashar, J Chaturvedi, "Design and Autonomous Control of the Active Adaptive Suspension System Rudra Mars Rover", available from

<http://arxiv.org/ftp/arxiv/papers/1407/1407.5197.pdf>

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- [2]. Rong Zhang, Christian Vogler, Dimitris Metaxas; Human Gait Recognition; IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, VOL. 27, NO. 11, NOVEMBER 2005.
- [3]. Ming-Gang Wen; Human Gait Classification Using Motion Information; Journal of Information Technology and Applications Vol. 1 No. 1, May, 2006, pp. 1- 8
- [4]. A. Hayder, J. Dargham, A. Chekima, and G. M. Ervin, "Person Identification Using Gait," International Journal of Computer and Electrical Engineering, Vol. 3, No. 4, August 2011.
- [5]. Jay Prakash Gupta, Pushkar Dixit, Nishant Singh, Vijay Bhaskar Semwal, "Analysis of Gait Pattern to Recognize the Human Activities," International Journal of Interactive Multimedia and Artificial Intelligence July 2014.

Questions?

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Thank you

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Have a great day
ahead

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